

Raúl Figueroa Sierra

Personal Information

Full name: Raúl Figueroa Sierra
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Research Interests

My research lies in mathematical logic, with primary interests in descriptive set theory, infinite combinatorics, and forcing, as well as their applications to topology and infinitary logic.

Education

Universidad de los Andes

Bogotá, Colombia

- Ph.D. in Mathematics (2019–2025)
Thesis: *Combinatorics of ideals on ω : the Category Dichotomy and the Measure Dichotomy*.
Advisors: Michael Hrušák (UNAM) and Carlos Di Prisco (Uniandes).
- M.Sc. in Mathematics (2017–2019)
Thesis: *Cardinal Quantifiers in Continuous Logic*.
Advisor: Xavier Caicedo Ferrer.

Universidad Nacional de Colombia

Bogotá, Colombia

- B.Sc. in Mathematics (2012–2017)
Thesis: *On Some Categorical Notions of Non-Elementary Classes*.
Advisor: Pedro Zambrano Ramírez.

Publications and preprints

- *The Category Dichotomy for Ideals*.
Joint with Alan Dow, Osvaldo Guzmán, and Michael Hrušák.
Annals of Pure and Applied Logic (2026).
- *Are scales Fréchet?*
Joint with Osvaldo Guzmán, Michael Hrušák, and Adam Kwela.
To appear in *Topology and its Applications*.
- *Compactness of continuous logic with generalized quantifiers*.
Joint with Xavier Caicedo.
In preparation.
- *Combinatorics of the Arkhangel'skii–Franklin space S_ω* .
In preparation.

Selected Talks

- *The optimal projective complexity of a counterexample to the Category Dichotomy.*
XXI Latin American Symposium on Mathematical Logic (SLALM), Bogotá, Colombia, June 1–5, 2026.
- *Ideales de subconjuntos densos en ninguna parte sobre espacios de Dow.*
XIII Simposio Nororiental de Matemáticas, Bucaramanga, Colombia, December 1–5, 2025.
- *Espacios topológicos y la Dicotomía de Categoría.*
Taller Mexicano de Topología, Mexico City, Mexico, June 15–19, 2025.
- *Pathological Lower Semicontinuous Submeasures and F_σ Ideals on ω* (Poster).
Young Set Theory Workshop, Münster, Germany, May 29 – June 3, 2023.
- *Compact Extensions of Continuous Logic and Partition Properties.*
XVIII Latin American Symposium on Mathematical Logic (SLALM), Concepción, Chile, 2019.

Teaching Experience

Universidad de los Andes, Departamento de Matemáticas

- **Adjunct Lecture** 2025-II – 2026
 - Teaching load: 12 hours/week.
 - Courses taught:
 - * 2026 (Summer course): Differential Calculus (MATE-1203).
 - * 2026-I: Differential and Integral Calculus (MATE-1254), Precalculus (MATE-1201).
 - * 2025-II: Linear Algebra I and Calculus III (MATE-1253), Precalculus (MATE-1201).
 - Responsibilities: course planning, lectures, exam preparation, grading, student advising.
- **Graduate Teaching Assistant** 2017-II – 2025-I
 - Teaching load: 6 hours/week.
 - Courses assisted: Linear Algebra, Differential Calculus, Integral Calculus and Differential Equations, Linear Algebra and Calculus III, Integral Calculus and Probability, Multivariable Calculus.
 - Duties: problem sessions, grading, academic support.
- **Research Assistant** June–July 2018, June–July 2019

Universidad Nacional de Colombia, Facultad de Ciencias

- **Teaching Assistant** 2014–2017
 - Basic Mathematics Program (March–June 2016; June–July 2016).
 - Faculty of Sciences (September–December 2014; February–June 2015).
 - Youth summer courses on mathematical outreach (March–May 2017).